

# Data Communication

## Content 10: Multiple Access

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# Access Control

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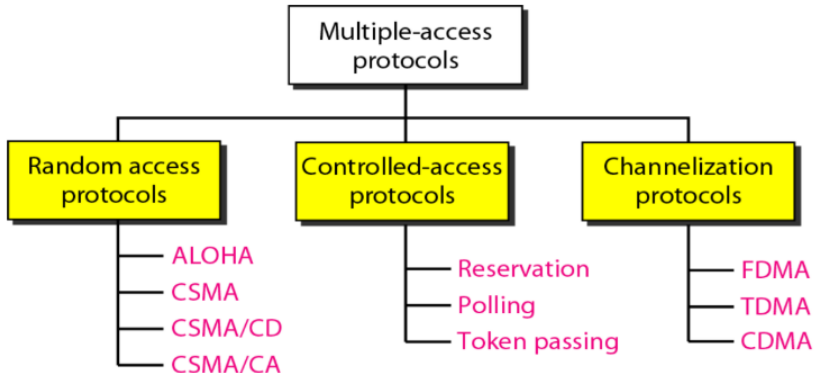
Data Link layer divided into two sub-layers:

- The upper sub-layer is responsible for data-link control
- The lower sub-layer is responsible for resolving access to the shared media



# Multiple Access Protocols

When nodes or stations are connected and use a common link, called a multipoint or broadcast link, a multiple-access protocol is needed to coordinate access to the link.



# Random Access

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- In random access or contention methods, no station is superior to another station and none is assigned the control over another.
- If more than one station tries to send, there is an access conflict – COLLISION – and the frames will be either destroyed or modified.
- To avoid access conflict, each station follows a procedure. The procedure includes instructions on:
  - When can the station access the medium?
  - What can the station do if the medium is busy?
  - How can the station determine the success or failure of the transmission?
  - What can the station do if there is an access conflict?



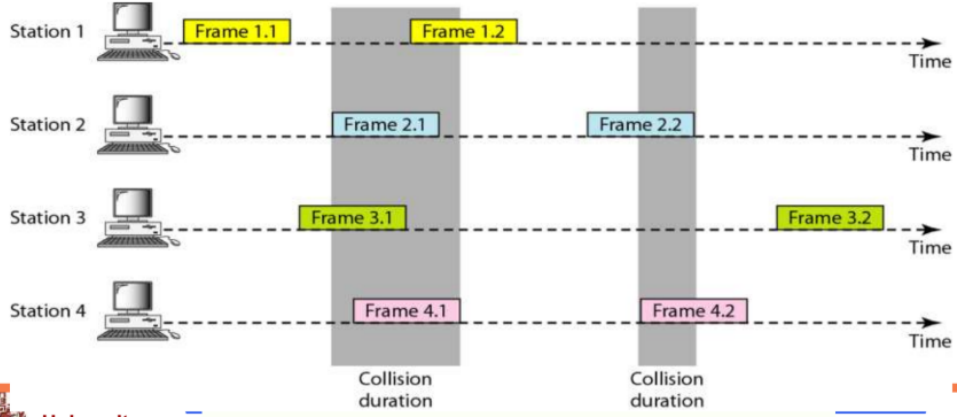
# Pure ALOHA

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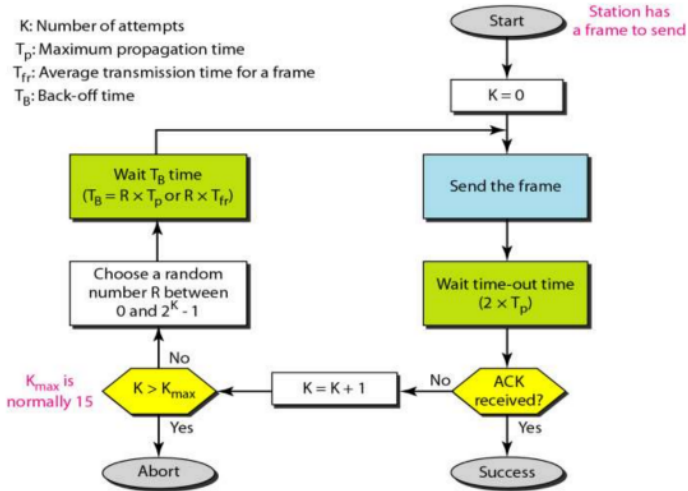
- Each station sends a frame whenever it has a frames to send.
- However, there is only one channel to share, there is the possibility of collision between frames from different stations.



# Pure ALOHA



# Pure ALOHA: Flowchart



# ALOHA Protocol Rule

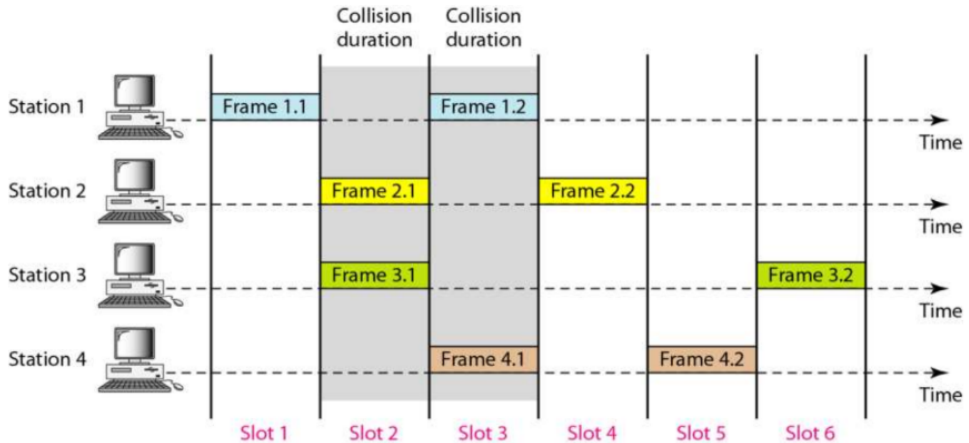
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- Each station sends a frame whenever it has a frames to send.
- After sending the frame, the station waits for an acknowledgment.
- If it does not receive an acknowledgement during the 2 times the maximum propagation delay ( $2 \times T_p$ ),
- It assumes that the frame is lost; it tries sending again after a random amount of time ( $T_B = R \times T_P$  or  $R \times T_{fr}$ )
  - $T_p$  (Maximum propagation time)= distance / propagation speed
  - $T_B$  (Back off time) : Random exponential back off time where  $R$  is random number choose between 0 to  $2^K - 1$  and  $K$  is the number of attempted unsuccessful transmissions
  - $T_{fr}$  (the average time required to send out a frame)



# Slotted ALOHA

We divide the time into slots of  $T_{fr}$  and force the station to send only at the beginning of the time slot.



# Carrier Sense Multiple Access (CSMA)

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- CSMA is based on the principle “sense before transmit” or “listen before talk”
- CSMA can reduce the possibility of collision, but it cannot eliminate it.
- The possibility of collision still exists because of propagation delay; a station may sense the medium and find it idle, only because the first bit sent by another station has not yet been received.



# Persistence Methods

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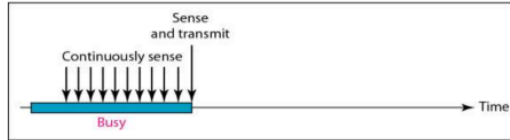
Persistence strategy defines the procedures for a station that senses a busy medium.

Three strategies have been developed:

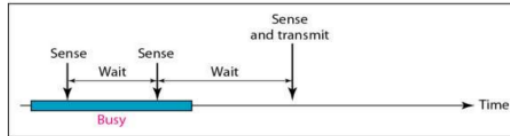
- **1-persistent Method:** After the station finds the line idle, it sends its frame immediately (with probability 1).
- **Non-persistent Method:** A station that has a frame to send senses the line. If the line is idle, it sends immediately. Otherwise, it waits a random amount of time and then senses the line again.
- **p-persistent Method:** After the station finds the line idle, with probability  $P$ , the station sends its frames. With probability  $q=1-p$ , the station waits for the beginning of the next time slot and checks the line again.



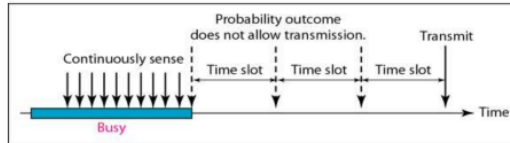
# Persistence Methods



a. 1-persistent

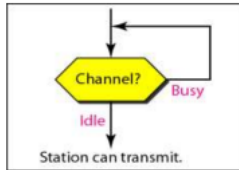


b. Nonpersistent

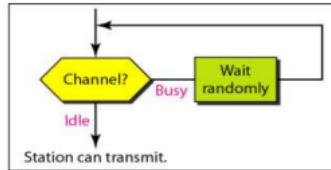


c. p-persistent

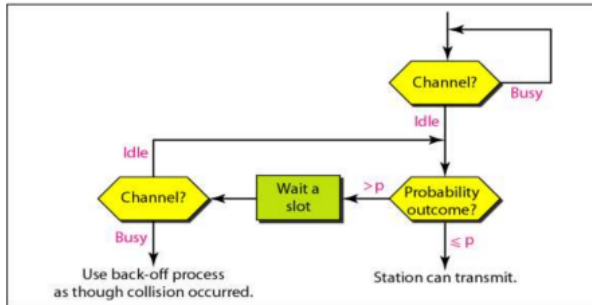
# Persistence Methods



a. 1-persistent



b. Nonpersistent



c. p-persistent

# CSMA with Collision Detection (CSMA/CD)

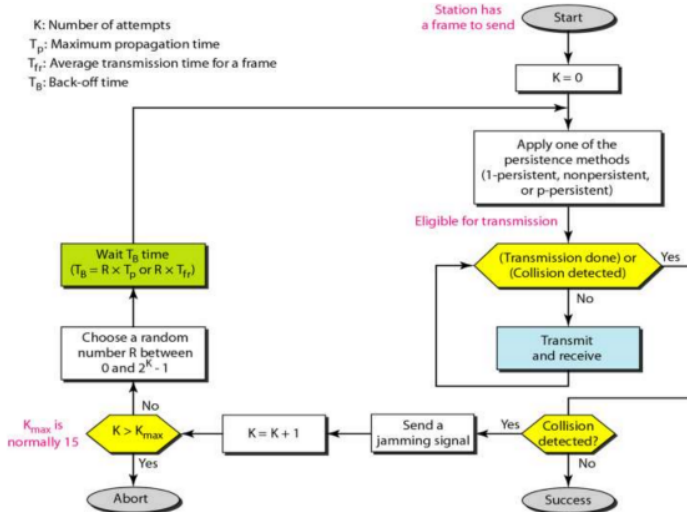
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CSMA/CD adds a procedure to handle the collision.

- A station monitors the medium after it sends a frame to see if the transmission was successful.
- If there is a collision, the frame is sent again.



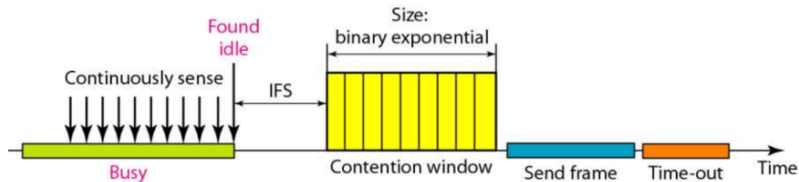
# CSMA with Collision Detection (CSMA/CD)



# CSMA with Collision Avoidance (CSMA/CA)

CSMA/CA was invented for the wireless networks to avoid collisions. Collisions are avoided through the use of CSMA/CA's three strategies:

- Interframe Space (IFS)
- Contention Window
- Acknowledgements



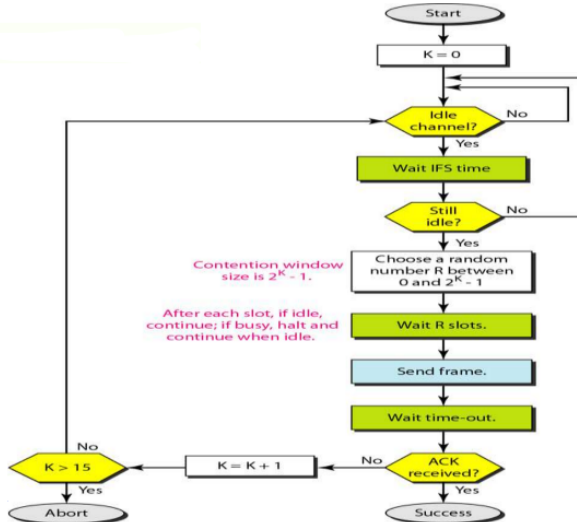
# CSMA with Collision Avoidance (CSMA/CA)

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- The IFS can also be used to define the priority of a station or a frame.
- If the station finds the channel busy, it does not restart the timer of the contention window; it stops the timer and restarts it when the channel becomes idle.

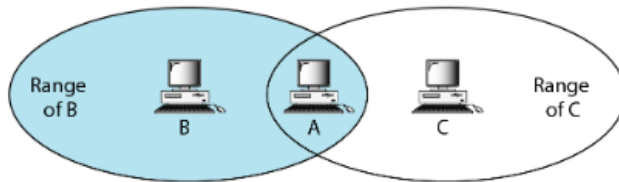


# CSMA with Collision Avoidance (CSMA/CA)



# Hidden Station Problem

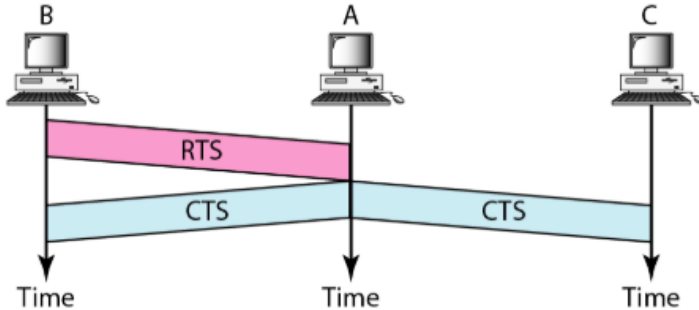
- Station B is sending data to station A.
- Station C also wants to send data to station A.
- Station C is out of B's range and cannot detect B's transmission.
- C assumes the medium is free and starts transmitting.
- A receives transmissions from both B and C, causing a collision.
- Therefore, stations B and C are hidden from each other with respect to A.



B and C are hidden from each other with respect to A.

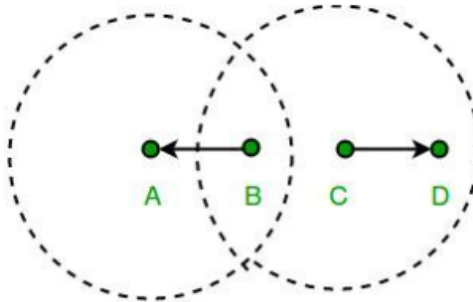
# Hidden Station Problem: Solution

The CTS frame in CSMA/CA handshake can prevent collision from a hidden station.



# Exposed Station Problem

- Station B is sending data to station A.
- Station C also wants to send data to station D.
- Station C can hear B's transmission and assumes that the medium is busy.
- However, C's transmission to D would not interfere with B's transmission to A.
- C unnecessarily remains silent, reducing channel utilization.
- Therefore, C is an exposed station with respect to B.



# Exposed Station Problem: Solution

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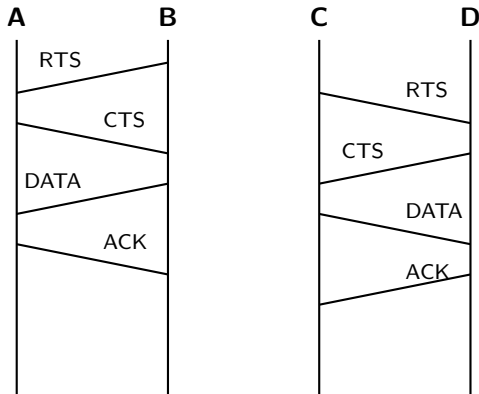
The RTS/CTS handshake can allow an exposed station to transmit without interfering with an ongoing transmission.

- B sends an RTS to A, and A responds with a CTS.
- C hears B's RTS but does not hear A's CTS.
- C can therefore initiate a transmission to D.
- B and C can transmit simultaneously because their receivers (A and D) are out of each other's interference range.

**RTS/CTS cannot properly solve the Exposed Station Problem.**



# Exposed Station Problem: Solution



# References

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- Data Communication and Networking (4th Ed) - Behrouz A. Forouzan  
Chapter 10: 10.1 - 10.5  
Chapter 14: 14.1



# Thank You

